Whenever possible, Caterpillar recommends the use of mechanical skis when encountering a high degree of roughness in the base to be paved. When using a mechanical ski and best paving practices, a 60 percent to 70 percent improvement in ride quality measurements should be realized when initial roughness is high. More information about paving techniques for smoothness is contained in Unit 7.



9-meter (30') multi-articulated, outboard leveler used to improve an extremely rough grade.

The Cat Fore 'N Aft Leveler is frequently installed when there is no suitable grade reference outside the paving width or when the paving width is so great that an outboard leveler cannot be installed. The Fore 'N Aft Leveler consists of one 2.5 m (8') aluminum beam with two articulated short beams. Each of the short beams has two articulated contact runners. So the front section is very similar to the outboard leveler. A post extends from the center of the main beam. A string extends from the post on the front section over the screed and connects to a post on a following beam behind the screed.



Cat 9-meter (30') Fore 'N Aft Leveler provides averaging capability inside the paving width.

The following beam with string post has an articulated pan that rides on the new bituminous layer. The string that extends between the two posts is 9 m (30') long. The center of the string has a roughness reduction factor similar to the outboard leveler.

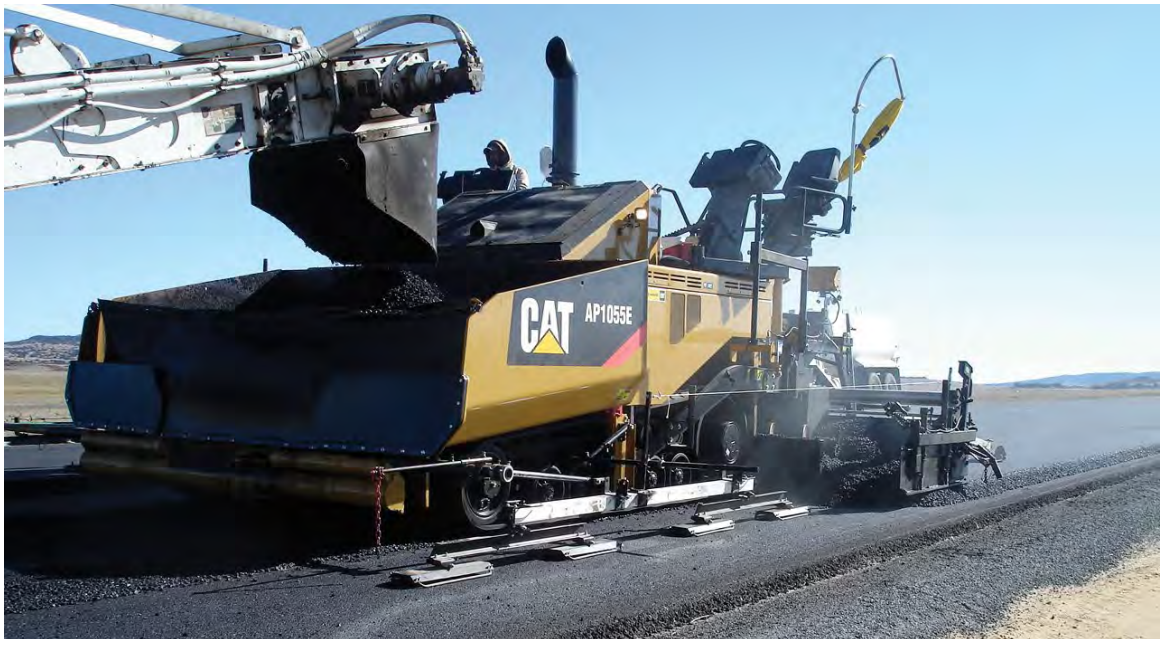


The rear section of the Fore 'N Aft Leveler rides on the new bituminous layer.

User Tip: *While heating the main screed plate and the extender screed plate, heat the drag pan on the trailing beam. Most crews use charcoal to heat the pan so it won't stick to the hot bituminous material. A propane torch can also be used for faster heating.*

Both types of Cat mechanical skis are towed by the tractor portion of the paver. The front tow arm has a collar that pivots around a shaft extending from the tractor steering guide arrangement. The tow arm always tows the ski; it never pushes the ski. The tow arm should be nearly parallel, or slightly nose up, to the beam of the ski.

The rear section of the Fore 'N Aft Leveler is attached to a shaft extending off the tractor frame. The rear section of the Outboard Leveler is attached to a shaft extending off the back of the screed. These connections are guide mechanisms that help keep the levelers straight.



The tractor tows the mechanical averaging ski.

The center of the stringline that spans the leveling devices is the point of maximum averaging. The most improvement in ride quality occurs when the center of the string is just behind the tow-point connection and the sensor is located above the center of the stringline.

The string itself should be multi-strand at least 3 mm (0.125") in diameter when using a sonic sensor. Do not use monofilament line or wire. They are poor targets for sonic sound pulses.

The string must be taut, especially when using a mechanical grade sensor. The sensor wand will always exert some downward or upward pressure on the string. The string has to be taut to avoid being bowed by the wand pressure.

The string should be at least 10 cm (4") above the body of the ski. If the sonic sensor loses the string as its target or if the mechanical sensor wand slips off the stringline, the Cat Grade Control System will lock to grade. That is to say, the system will ignore any sudden grade deviation that is in excess of 5 cm (2").



Mechanical or sonic sensors have equal accuracy using the ski stringline as a guide.



Cat sonic grade sensors are oriented horizontally when reading a stringline.

Use the Sensor Select function on the display to select the sonic sensor with stringline or mechanical sensor. Turn the sonic sensor horizontally to the stringline and center the sensor over the stringline. Always have the sensor cable connection on the left side of the sensor. Having the cable on the left side enables the outer transducers to act as guides for sensor location over the string. The display will indicate to the operator if the sonic sensor is drifting right or left over the stringline.

Mechanical, contact-type averaging skis are preferred for some applications rather than electronic, non-contact averaging skis. Some of the reasons to choose mechanical averaging skis are:

- **Maximum averaging needed.** A mechanical ski is the better choice when there is a high degree of roughness in the grade to be paved. Remember, a mechanical ski like the multi-articulated outboard leveler has a predictable averaging factor of 88 percent.
- **No time constraints.** When there are no time restrictions on a project, mechanical skis can be used. Typically, those projects have long passes in one direction without the need to set back and restart. Those projects are typically day shifts in rural areas.
- **Difficult specifications.** When the threshold for earning the highest level of ride quality incentive compensation is difficult to attain, choose mechanical averaging skis whenever possible. An example would be whenever there is a need to be under 0.6 meters per kilometer (40" per mile) using the International Roughness Index. Or, when using Profile Index with zero blanking band and there is a need to be under 16 centimeters per kilometer (10" per mile).



Mechanical ski is the better choice when there is a high degree of roughness in the grade to be paved.



Cat Paver equipped with Trimble three-dimensional guidance system.

[THREE DIMENSIONAL PAVING]

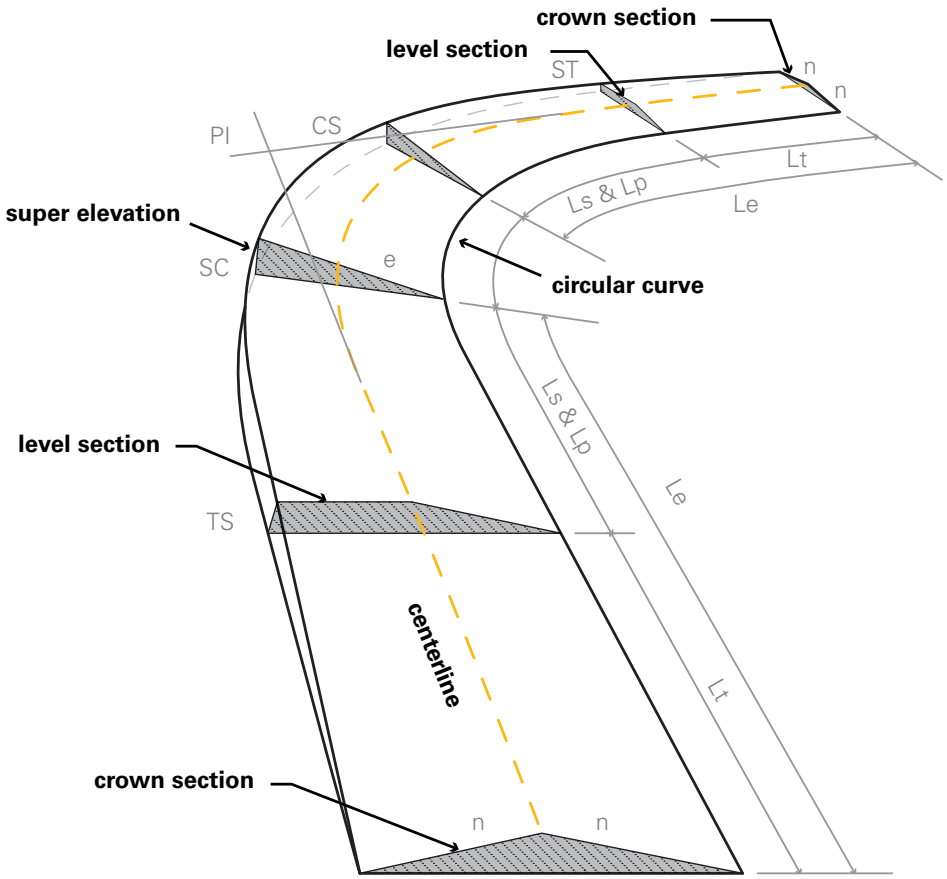
The material in Unit 6 to this point has referred to on-machine grade and slope control. These types of guidance systems are sometimes referred to as two-dimensional systems. They control the screed based on measurements of the grade or slope as paving progresses. Another type of guidance system is available. Systems referred to as three-dimensional receive the input that controls screed reaction from sources that are independent of the paver.

A wide range of construction equipment uses three-dimensional guidance systems. This list of applications ranges from initial phases of surveying, clearing and earthwork to the preparation and compaction of the final aggregate base to paving and finally to pavement maintenance.

On a Cat Paver, the Trimble PCS900 provides an upgrade to Trimble or Cat control systems already in place on the paver.



A Universal Total Station provides off-machine guidance feedback for the paver.



Universal Total Stations perform stake-less surveys and can create complicated designs.

To provide the off-machine guidance that is necessary for three-dimensional paving, there must be at least one Universal Total Station on the project. For extreme accuracy (+/- 1.75 mm / 0.07"), stations should be within 350 m (1,150') of the paver. They provide continuous design feedback to receivers on the paver. The information can relate to a design that was created through original surveys done by the station or to a map of the existing structure that was created by a cold planer, for example.

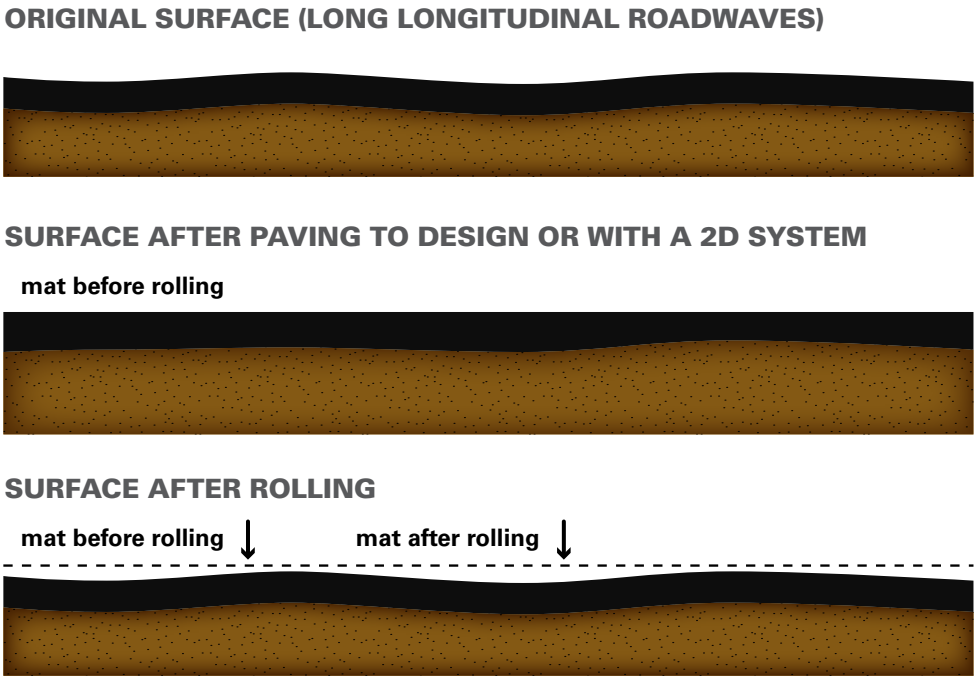
Design information created and stored by the three-dimensional system can be transmitted to construction equipment during every phase of the project. From a paving standpoint, off-machine control simplifies working on difficult segments of a project—areas like super-elevations. The responsibility for looking ahead and reacting to slope changes on grade stakes is taken away from the operators and is done according to the design data and measurements transmitted by the Universal Total Station. Additionally, if all phases of earthwork and base preparation have been done using equipment under three-dimensional guidance, the final step, paving with expensive bituminous material will be done to precise yield calculations.

Machine targets are attached to masts that extend above the paver. The position of the targets is an indication of the position of the screed. This information is continuously sent to the control module. The control module signals the tow-point cylinder valves to move up or down to keep the screed at the design elevation and slope. The ability to integrate existing grade conditions with the design criteria gives three-dimensional guidance systems a significant advantage over two-dimensional systems. That advantage is the ability to create a layer that compensates for the rate of compaction in the layer.





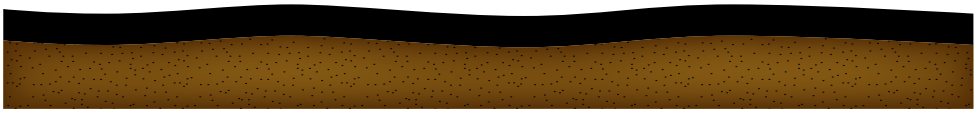
Machine targets on the paver are tracked by the Universal Total Stations.



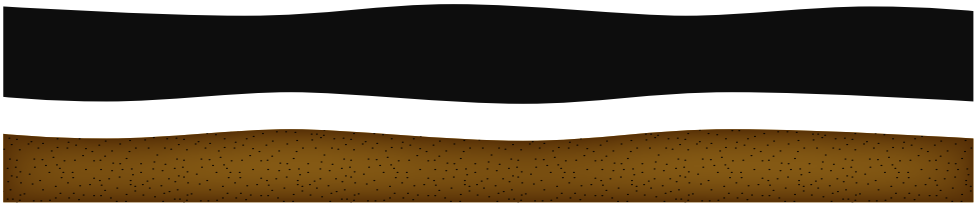
Using conventional two-dimensional guidance systems, it is possible to create an extremely smooth surface on a layer that is placed over an irregular grade. Using averaging skis especially, it is common to fill in low spots and level off high spots in the grade being paved. A wavy road can be made smooth by screed-laid material. However, due to the compaction rate – about 6 mm per 25 mm (0.25" per inch)—the layer with variable thickness experiences a variable compaction rate.

Consequently, the roughness of the original grade returns to a certain extent after compaction. The surface is smoother, but it can be even better.

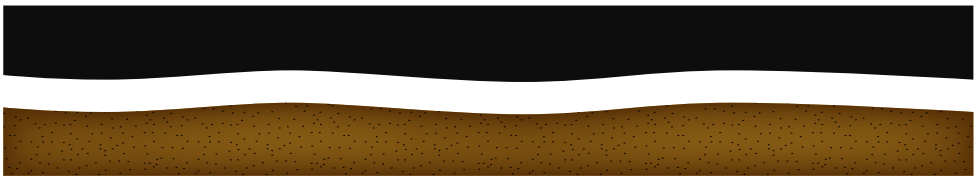
ORIGINAL SURFACE (LONG LONGITUDINAL ROADWAVES)



3D PAVED SURFACE (USING THE DIFFERENTIAL DESIGN)



3D PAVED AND COMPACTED SURFACE (DIFFERENTIAL DESIGN)



A three-dimensional control system can accept the compaction factor (in this case 1.25) and plan for the compaction factor in the design. The screed will be directed to increase thickness even more when paving over low spots and to decrease thickness when paving over high spots in the grade. In this example, the differential compaction will produce a smoother, more level surface.

Many types of projects are candidates for three-dimensional paving. Airport runways and aprons that need precise elevation control are already being completed with three-dimensional guidance systems. Large highway projects, both new construction and repair, benefit from close control of yields and improved smoothness. Contractors doing large site development projects are finding that survey work and machine control with three-dimensional systems save time and reduce costs.

[SUMMARY]

Automatic grade and slope control on pavers is an option designed to make the crew's jobs easier and to improve the quality of the final product. A crew that is well-trained in the uses of grade and slope control can set up the paver to be successful on all applications. Crewmembers must know the important considerations on any given project, and then should be able to decide what type of screed response is needed for the application. They should know how sensor position affects screed reaction. They should be able to select and install averaging skis that are correct for the application.

Remember, the use of grade and slope control cannot overcome defects caused by the crew violating fundamental paving practices or by making big mistakes. Make sure the fundamentals are done correctly every time and do not take shortcuts.


CAT
AP1055F
1096
GRADE CONTROL



Unit 7
PAVING APPLICATIONS

While each application presents its own challenges, paving fundamentals typically remain constant. The use of automatic grade and slope control is one of the few key success factors that changes with almost every application. Grade Control and other technological advances have helped contractors become more successful while handling challenging worksites and major applications.



Unit 7 focuses on successful techniques used in a variety of paving applications. Each application presents its own challenges and some require the use of special options or equipment. In general, however, the fundamentals of paving are the same for every application. Likewise, the need to be efficient and to avoid mistakes is the same for every paving application. However, one important factor that changes with almost every application is the use of automatic grade and slope control. Therefore, setting up and using grade and slope control systems are focal points covered in this unit.

Not every paving application or variable can be addressed in a publication on paving. As much as possible, this unit will cover the major applications, as well as those that typically are the most challenging. The general topics that are found in this unit include:

- Butt joint preparation
- Paving for yield (thicknesss)
- Paving for smoothness
- Paving for transverse profile (slope)
- Paving to elevations
- Longitudinal joint construction
- Paving super-elevations
- Truck exchange procedures
- Windrow paving techniques
- Paving with material transfer vehicles
- Balancing production (paving speeds)
- Echelon (staggered) paving
- Roller compacted concrete
- Paving aggregate base

[BUTT JOINT PREPARATION]

A butt joint is commonly defined as the transverse intersection of a previously laid and compacted bituminous layer with a new layer of hot or warm bituminous material. Butt joints, usually in conjunction with some sort of thin boards or metal strips are often the starting point of a paving pass. Too frequently, the transverse butt joint is an area of significant roughness and isolated bumps that requires grinding to meet smoothness or bump-control guidelines. The ultimate goal of transverse joint construction and compaction is to build a joint that is smooth and durable.

A large part of proper transverse joint construction is preparing the paver. Unit 4 covers preparing to pave with a vibratory screed. Unit 5 covers preparing to pave with a tamping and vibratory screed.

Following are other steps that will help assure that the butt joint is ready to be part of the starting reference for the paver.



The end of a pass is prepared, so creating a butt joint is easy when paving resumes.

When the crew completes the paving pass at the end of a shift or at the end of a designated distance, leaving enough material in the auger chamber is a good practice, so the screed maintains its elevation until the paver is stopped. When the operator picks up the screed and pulls forward, the head of material that was in the auger chamber should be uniform in size across the width of the layer. The crew should then put down some non-stick material and rake out the remaining asphalt over the ground cover, building a tapered ramp for the compaction equipment and for traffic, if any.

If the crew has a habit of using most of the material in the auger chamber before stopping at the end of the pass, the screed will lose elevation. The surface of the bituminous layer at the end of the pass will have a downward slant. If the butt joint is located within the area of downward slant, the screed will take off with a downward direction and the layer will be too thin.



Mark the location of the butt joint where the layer is flat and not slanting downward.

Check the condition of the compacted layer with a slope board or electronic slope meter when preparing to pave. Find a location where the layer is flat and mark that location. Remove the material in front of that mark.

User Tip: *One way to help find where the layer starts to slant downward at the end of the pass is to paint a mark on the layer at the point where the crew switches the feeder system from Auto mode to Manual mode. Screed operators commonly use manual feed to control the material in front of the screed at the end of the pass. As soon as they switch to the Manual mode, the forces in front of the screed begin to change and screed elevation becomes variable.*

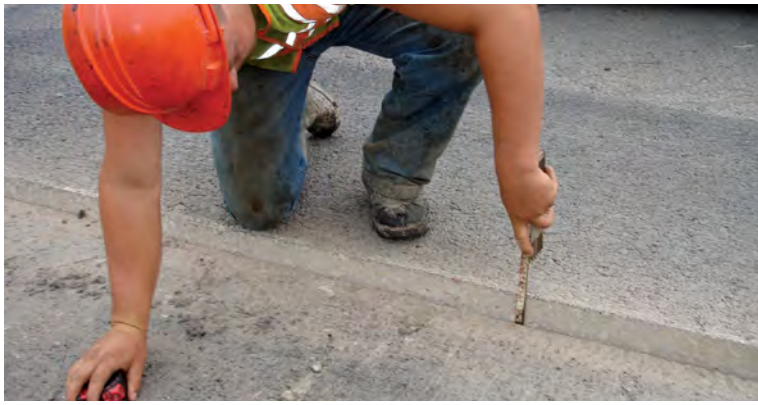
Cut a straight face for the butt joint. A rotary saw with adjustable depth control is the most common device. Use a skid steer or small loader to remove the material in front of the butt joint. Sweep the area in front of the butt joint and a short distance back from the butt joint.



A butt joint should have a straight, vertical face.

Note: *Never take off from a joint having a rounded face. In some applications, a cold planer works immediately in front of the paver. After the cold planer makes its plunge cut, the face of the butt joint forms an arc. Take the time to vertically cut the face with a saw before paving resumes.*

Most crews use some sort of thin wooden boards or metal strips to create the correct pre-compaction height at the transverse joint. For mats laid by a vibratory screed, the rule of thumb is 6 mm (0.25") compaction for every 25 mm (1") of uncompacted thickness. For mats laid by a tamping screed, the assumption is 1 cm (0.4") compaction for every 5 cm (2") of uncompacted thickness. Those guidelines are fairly accurate, but are affected by the type of bituminous material being placed, the temperature of the material, and the type of screed being used. To be even more precise when selecting the thickness of the starter boards, measure the height of the vertical joint. Subtract that dimension from the thickness of the intended layer. The difference between those two dimensions is the compaction rate and the exact thickness of the required starter boards.



Measure the height of the butt joint to help select the correct starter board thickness.

Some crews are successful starting with the screed sitting directly on the bituminous layer about 1 m (3') behind the face of the butt joint and bringing the screed up slightly as the paver begins to pave to create the correct pre-compaction thickness at the transverse joint. Caterpillar recommends use of this technique only for relatively thin layers—no more than 5 cm (2") thick. Thicker layers require the assistance of starter boards at the butt joint.

Pay special attention to tack on the face of the vertical butt joint and the area just in front of the transverse joint. The tack truck with spray bar may not be able to coat the face of the joint completely. When using a spray wand or hand applicator, be sure the butt joint is thoroughly coated with tack before locating the paver over the joint.

If the crew has prepared the joint and the paver properly, the joint should need only minimal handwork prior to compaction.



Use a spray wand or hand applicator to apply tack to the face of the butt joint.



A properly built transverse butt joint requires minimal handwork prior to compaction.

[PAVING FOR YIELD]

Yield is the ratio of the weight of the actual material being placed versus the weight of material that has been estimated in a bid or contract based on length, width and depth. For example, if a project bid called for installation of 5 000 tonnes (5,500 tons) of material and the actual usage is 5 200 tonnes (5,720 tons), the calculated yield is:

$$\frac{5200\,T}{5000\,T} = 104\%\ \text{Yield}$$

Another common way to think of yield is in terms of length that can be paved by a certain amount of material when the weight of the material is known, as well as the planned paving width and paving depth. In that case, yield is the ratio of the distance actually paved over the calculated paving distance.

Consider this example. The compacted weight of the material is 2450 kg/m³ (153 lb/ft³). The compacted depth is 5 cm (2") at a width of 3.66 m (12'). If the average haul unit carries 18 tonnes (19.8 tons) and the distance covered by one truckload is 41 m (135'), what is the yield of that truckload of material? This calculation is more complicated. The simplest way to handle this calculation is to use the Paving Production Calculator or similar software.

YIELD CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[1.97] *in*	[50.0] *mm*
Paving Width:	[12.00] *feet*	[3.658] *meter*
Material Density Uncompacted:	[153] *lbs/ft³*	[2450] *kg/m³*
Truck Capacity or Total Tonnage:	[19.8] *ton*	[18.0] *tonnes*
Length of Mat at 100% Yield:	[131.38] *feet*	[40] *meter*
Actual Length of Mat produced:	[134.51] *feet*	[41] *meter*
% Yield for given truck load or tonnage:		[102]

Thickness:	[1.97] *in*	[50.0] *mm*
Length of Mat produced:	[134.51] *feet*	[41] *meters*
Width:	[12] *feet*	[3.658] *meters*

Exit

Using the Paving Production Calculator, yield can be calculated for the distance that one truckload of material will cover.

In this example, the actual distance covered by one average truckload of material is 41 m (135'). The calculated distance is 40 m (131'). The yield is 102 percent in this example. The same yield calculation can be prepared for the distance covered in one shift.

YIELD CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal
Exit

General Inputs	Imperial Units	Metric Units
Paving Thickness:	[1.97] in	[50.0] mm
Paving Width:	[12.00] feet	[3.658] meter
Material Density Uncompacted:	[153] lbs/ft³	[2450] kg/m³
Truck Capacity or Total Tonnage:	[2204.6] ton	[2000.0] tonnes
Length of Mat at 100% Yield:	[14628.58] feet	[4459] meter
Actual Length of Mat produced:	[14435.70] feet	[4400] meter
% Yield for given truck load or tonnage:		[99]
	[80]	

Thickness:	[1.97] in	[50.0] mm
Length of Mat produced:	[14435.70] feet	[4400] meters
Width:	[12] feet	[3.658] meters

Yield can be calculated based on the weight of material being placed in one complete shift.

Substitute total tonnes placed during the shift for tons carried by one haul unit. In one shift, for example, the crew has installed 2 000 tonnes (2,200 tons) of bituminous material. That weight should cover a distance of 4 459 meters (14,629'). That weight actually covered only 4 400 meters (14,435'). Therefore, the yield is 99 percent for that shift.

Length of paving and width of paving are easily controlled on most projects. The variable that most often affects yield is paving thickness. In most situations, the thickness of the bituminous layer must be closely controlled, if meeting precise yield goals is required.

In order to closely control the thickness of the bituminous layer, the crew has to consider several variables. First, is the transverse profile (slope) correct? If the slope is not correct and the crew is responsible for correcting the transverse profile, then it is very likely that yields will be incorrect. This is a very important point.



Paving with automatic slope control to correct the transverse profile created a thin layer of material on the right edge, resulting in incorrect yields.

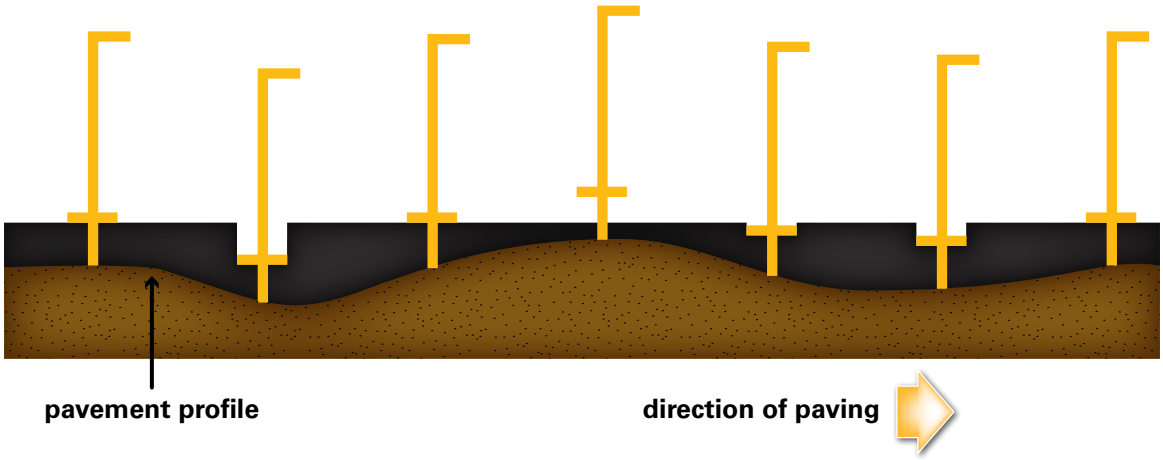
Note: *Anytime that the transverse profile is being corrected by the paver, the project owner should be informed that yields will probably be incorrect.*

On another project, assume that the transverse profile is correct. There is, however, roughness in the grade to be paved, and the crew is being asked to improve ride quality. In that situation, the crew has decided to install averaging skis.

If averaging skis are making substantial improvements in the ride quality of the structure, there will be some yield variation. However, it is possible that at the end of a shift, the high spots and low spots will average out and the daily yield may be close to the calculated yield. The same can be said when the crew controls the thickness of the layer manually.

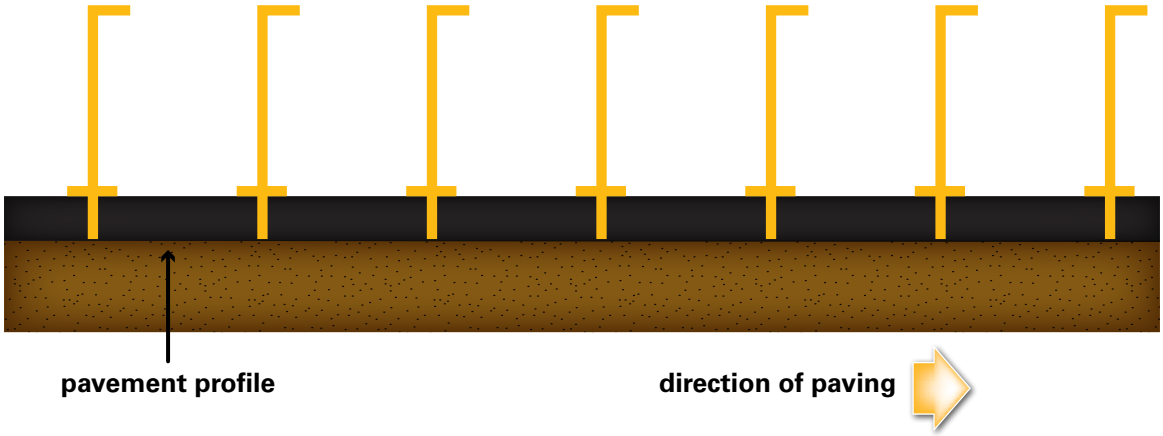


When averaging skis are used, the screed tends to fill in low spots and level off high spots with yields varying accordingly.



A series of depth measurements may show considerable variation when the screed is being controlled manually or if an averaging ski is installed.

Remember, the floating screed by its nature does not follow deviations in the grade. Rather, the floating screed, even without input from an averaging ski fills in the low spots and levels off the high spots. A series of depth measurements may show the thickness is too great in some areas and not thick enough in other areas. Do not make adjustments to thickness based on this series of measurements. Only make corrections when the layer is consistently thick or thin.



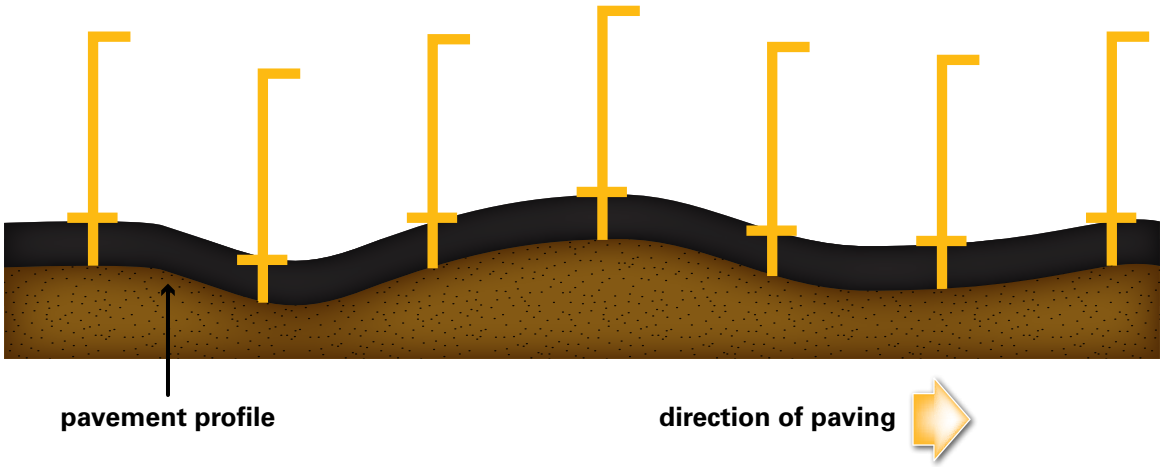
When the longitudinal profile is flat, it is easier to create a layer with uniform thickness.

If the grade is consistent over the bituminous layer being placed, it is easy to control thickness and yield. However, if the base has variations, there is only one way to control yield—pave with automatic grade control and a single sensor.



A single grade sensor installed close to the auger chamber provides the best control of layer thickness and yield.

The principles of Grade and Slope control are detailed in Unit 6. One of the principles dealt with how sensor position affects screed reaction. In order for the screed to follow the contours of the grade, position a single grade sensor adjacent to the auger chamber. Any grade deviation measured by the grade sensor will be transferred as a movement by the tow-point that is four times greater than the size of the grade deviation. The screed will react quickly. The layer thickness will change at exactly the location of the deviation—not good for smoothness, but great for yield.



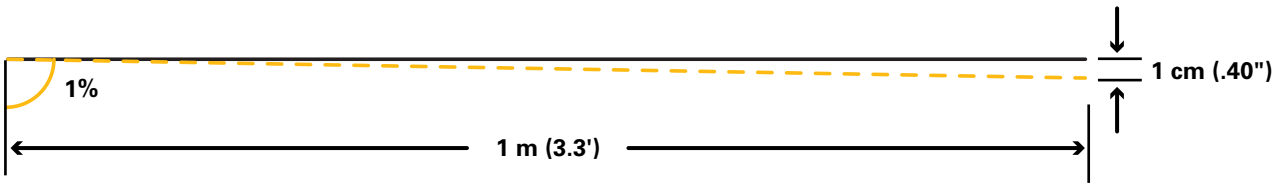
A screed that reacts quickly to grade deviations will create precise layer thickness and yield control.

The drawback to focusing on yield may be poor ride quality. If the longitudinal profile is rough and the Grade Controls are setup to control thickness, it is likely that longitudinal roughness will continue. If yield and layer thickness are the most important factors on a project, other factors may suffer.

[CREATING TRANSVERSE PROFILE (SLOPE)]

On many projects, the transverse profile or surface slope, of the bituminous layer is an important consideration. The slope of the structure accomplishes two important factors. One, slope controls the flow of water in the correct direction and to drain inlets, so moisture does not accumulate on the surface. Two, the transverse profile improves safety on curves (super-elevations).

If the longitudinal profile is rough and the Grade Controls are setup to control thickness, it is likely that longitudinal roughness will continue. The formula for calculating thickness changes when slope control is used to correct the transverse profile as described in Unit 6.



If the surface of the bituminous layer has a different slope than the base, the thickness of the layer will vary across the width of the layer. For every 1 percent of slope correction, thickness varies 1 centimeter per meter (0.125" per foot). To determine if the crew will use slope control and to determine whether slope will be controlled manually or automatically, check the slope of the base at numerous locations on the project.



Verifying transverse profile before paving helps the crew understand how to set up the paver.

If the application is placing the first base layer on granular material, check the transverse profile closely. Even if the grade trimmer or motor grader that finished the granular base material used some sort of automatic slope control, never assume that the slopes are correct. If discrepancies are found, notify the project owner or the engineer. They may want to correct the granular base before placing the more costly bituminous material. If discrepancies are found and paving must continue, notify the owner or engineer that the yields may be incorrect due to variable layer thickness caused by slope corrections.

Note: *In some areas, the use of slope control on the paver is required during installation of the first base layer on granular material. It is still a good idea to determine the accuracy of the transverse profile prior to paving.*

If the slope inspection reveals that slope corrections will be limited to 1 percent or less, automatic slope control can be used with confidence.

Set up the Cat Grade Control System for automatic operation after completing the steps to prepare the paver at the starting reference. To select slope control, tap the system status icon on the interactive display. The list of available sensors will show on the display. Tap the slope icon and depress the OK button.



Select slope control for the appropriate side of the paver.

Benchmark (null) the slope sensor by pressing and holding the Benchmark button (middle button) until the green "On Grade" light is illuminated. The system is in the manual mode at this time. Press and release the Mode Select button (second from bottom) and the word "AUTO" will appear in the system status area.



The slope value and slope direction shown on the display may or may not be correct at this time.

Check the slope of the screed while the screed is resting on the starting reference. Or, check the slope of the bituminous layer after the paver has pulled off the starting reference and the screed has reached equilibrium.



Verify the actual slope of the screed.

To calibrate the slope sensor, select the target slope value on the interactive display. Using the pop-up screen, enter the just measured slope value and slope direction. Depress the OK button to accept the value change and to return to the normal operating display.

By moving the appropriate tow-point cylinder up or down, automatic slope control will keep the slope of the layer within 0.05 percent of the benchmark value, if the deadband tolerance is not changed. The deadband can be set as low as 0.01 percent or as high as 0.50 percent.

To increase or decrease the slope benchmark while paving, depress the Up or Down arrow keys. Depressing and releasing an Up or Down arrow key changes the slope by 0.10 percent each time.



Calibrate the slope sensor with the measured slope.

Note: *The unit of measure for slope control can be changed to 0.01 percent, if desired.*

Note: *The Cat Grade Control System has a unique feature called "cross coupling." When the cross coupling feature is set at the default value of 100 percent, the grade-side tow-point and the slope-side tow-point move at the same time for extremely accurate slope creation. If the default value is lowered, the slope sensor must sense the slope change and signal the tow-point valve independent of the grade-side tow-point. Caterpillar recommends keeping the default value for cross coupling unless the grade sensor is referencing an extremely rough surface.*

Some applications may require slope corrections that are greater than 1 percent. For those applications, it may be beneficial to control slope manually. Remember, the thickness of the layer changes as the difference between the desired slope of surface and the actual slope of the grade becomes larger or smaller. Since the tow-point makes large moves when automatic slope control is active, it is likely that the tow-point cylinder will be fully extended or fully retracted before the slope correction is complete. When the tow-point cylinder has no more stroke available, the screed operator has to "gain back cylinder." (See pages 167 and 168 in Unit 6.) If large tow-point movements are expected, manual slope corrections can be made to avoid losing tow-point cylinder movement.



The display serves as a slope monitoring system at all times.

The display used in the Cat Grade Control System constantly shows the actual slope value in Measured Values area when slope is selected. Slope is also shown in the text bar at the bottom of the display when the text bar is configured to show slope. Slope shows on the text bar no matter what other machine conditions have been selected, as long as the slope sensor is connected. Grade control can be used on either side in the AUTO or MANUAL mode. The slope value will not be affected and serves as a constant reference for manual slope control. Be sure to verify that the slope sensor is calibrated.

The most common way to control slope manually is to use the depth control screws that are features of vibratory screeds.

Note: *Other grade and slope control systems have similar capability to monitor slope. Consult instructions included with other types of grade and slope control systems.*

MANUALLY DECREASING SLOPE



Assume that the system has been set up to control grade on the left side of the screed and that slope is to be maintained manually from left to right between 1.8 and 2.2 percent. Monitor slope using the right display. If the slope value on the display goes above 2.2 percent, layer thickness will need to be increased on the right side. Slowly turn the right depth-control screw in the direction of increase. Watch the slope value in the display. When the slope value starts to decrease, stop turning the depth-control screw. Remember, depth changes occur slowly over a distance equal to five tow arm lengths, and the slope value may continue to decrease. To stop the slope change, turn the depth-control screw in the opposite direction until no resistance is felt. At that point, the screed is at equilibrium and is floating without changing elevation (thickness).

Under the same conditions, turn the depth-control screw in the direction to decrease layer thickness and, therefore, increase slope from left to right.

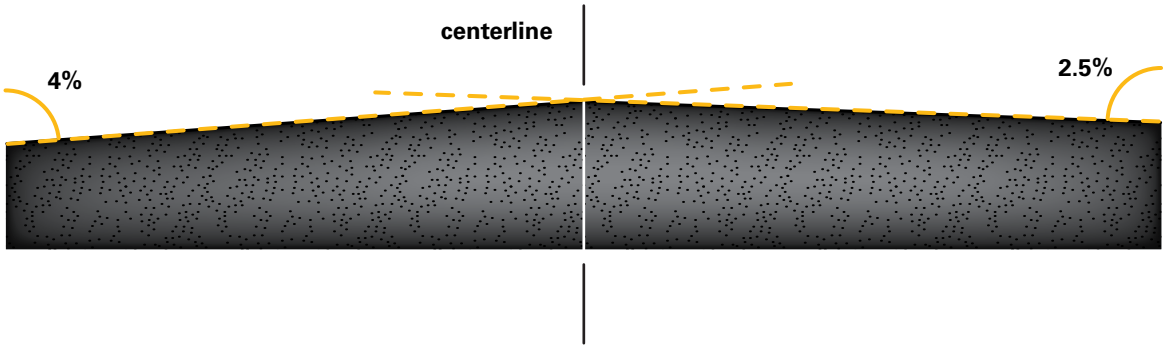
MANUALLY INCREASING SLOPE



User Tip: *Screed operators have to be alert for several conditions when making slope changes manually or automatically. When increasing layer thickness to correct the slope, the feeder system may tend to run faster due to the increased demand for material. The paver operator may have to make frequent adjustments to the ratio control dials to maintain the correct auger speed. Conversely, decreasing layer thickness to change slope, the feeder system may slow down too much. Or, more importantly, the screed may begin to drag aggregates, if the layer thickness is too thin to support the floating screed.*

Slope in conjunction with screed crown can be used to pave different slopes on either side of the structure's center line.

STRUCTURE WITH DUAL SLOPES



Some structures have slope left and slope right extending from a centerline. Sometimes the slope to the left and the slope to the right are not the same. The tendency would be to pave the left side of the structure in one pass creating the desired slope. Then, pave the right side of the structure at its specified slope. It is possible, however to pave this structure in one wide pass by simultaneously crowning the screed and sloping the screed. Setting up for this application requires some basic mathematics.

First, add the two slope percentages:

4% + 2.5% = 6.5%

Next, divide the sum of the slopes by 2:

$$\frac{6.5\%}{2} = 3.25\%$$

The result: The amount of crown in the main screed should be set to 3.25%.